IruBasha: Bilingual Vocabulary Learning Platform

IruBasha is a web-based bilingual vocabulary learning platform designed for Sri Lankan nursery and primary school students aged 3–12. It supports English, Sinhala, and Tamil, enabling children to translate words, listen to correct pronunciation, record their own pronunciation, and practice vocabulary through interactive quizzes.



Presented by: Team 06



Why IruBasha?

Sri Lanka is a trilingual country, yet early bilingual language learning remains challenging for young children due to limitations in traditional methods and existing global applications.

Problems identified

- Traditional Learning Methods are not Engaging
- Lack of Proper Sinhala and Tamil Digital Support
- Global Apps are not Child-centric or Localized

Gaps in Existing Solutions

- Lack of Local Language Accuracy
- No Pronunciation Practice for children
- Not Child-Friendly Gamification

IruBasha solves these problems by providing a child-friendly, localized, interactive, and web-accessible solution.

Existing Major Apps



duolingo



Google Translate

Made with GAMMA

How IruBasha is Different

Solution 1: Localized Bilingual Learning

- English <-> Sinhala <-> Tamil
- Accurate scripts
- Curated word base

Solution 2 : Rule-Based Intelligence with Prolog

- Instead of relying only on machine translation
- We use logic-based rules for known vocabulary
- Ensures accuracy for nursery-level words.

Solution 3: Pronunciation Practice

1. Audio Playback
2. Child records their own voice
3. Encourages speaking,not just reading

Solution 4: Gamification

- Quizzes
- Daily streaks
- Word mastery levels
- Progress statistics

The product is intentionally designed to be simple, guided, and self-explanatory for young learners.



Requirements Overview

Requirements define what the system must do, ensuring alignment with user needs and academic objectives. These were identified through observation of children's learning behavior, feedback from parents and teachers, and analysis of similar educational applications.

User Needs

Understanding the specific requirements of young learners.

Academic Objectives

Meeting educational goals for early language acquisition.

Application Analysis

Learning from existing educational tools and best practices.

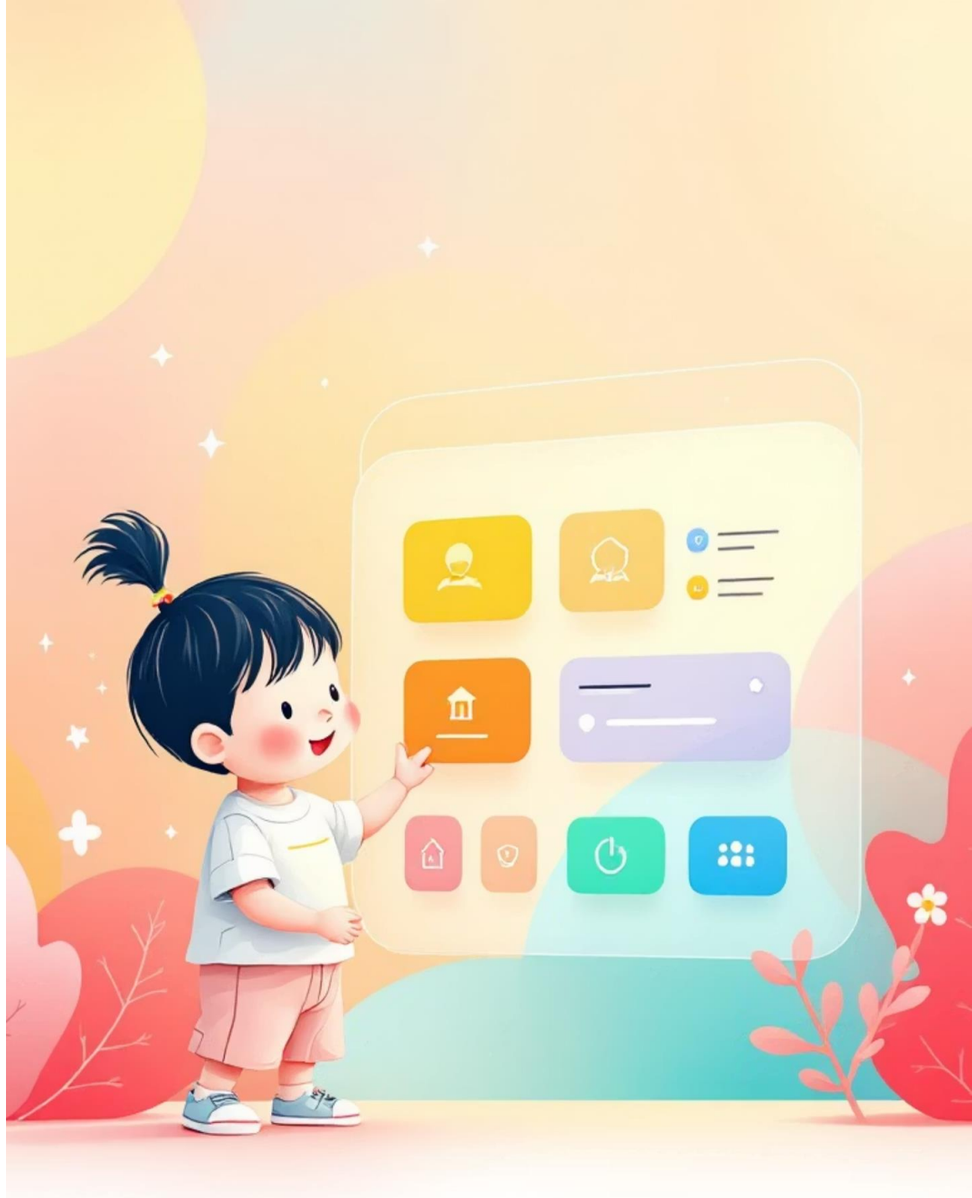
User Requirements

Functional User Requirements

- Enter an English word
- View Sinhala and Tamil translations
- Listen to correct pronunciation
- Record and replay own pronunciation
- Attempt vocabulary quizzes
- Save learned words in a personal word bank

Non-Functional User Requirements

- Easy to use for children aged 3–12
- Visually attractive and child-friendly
- Accessible via a web browser
- Fully responsive across devices



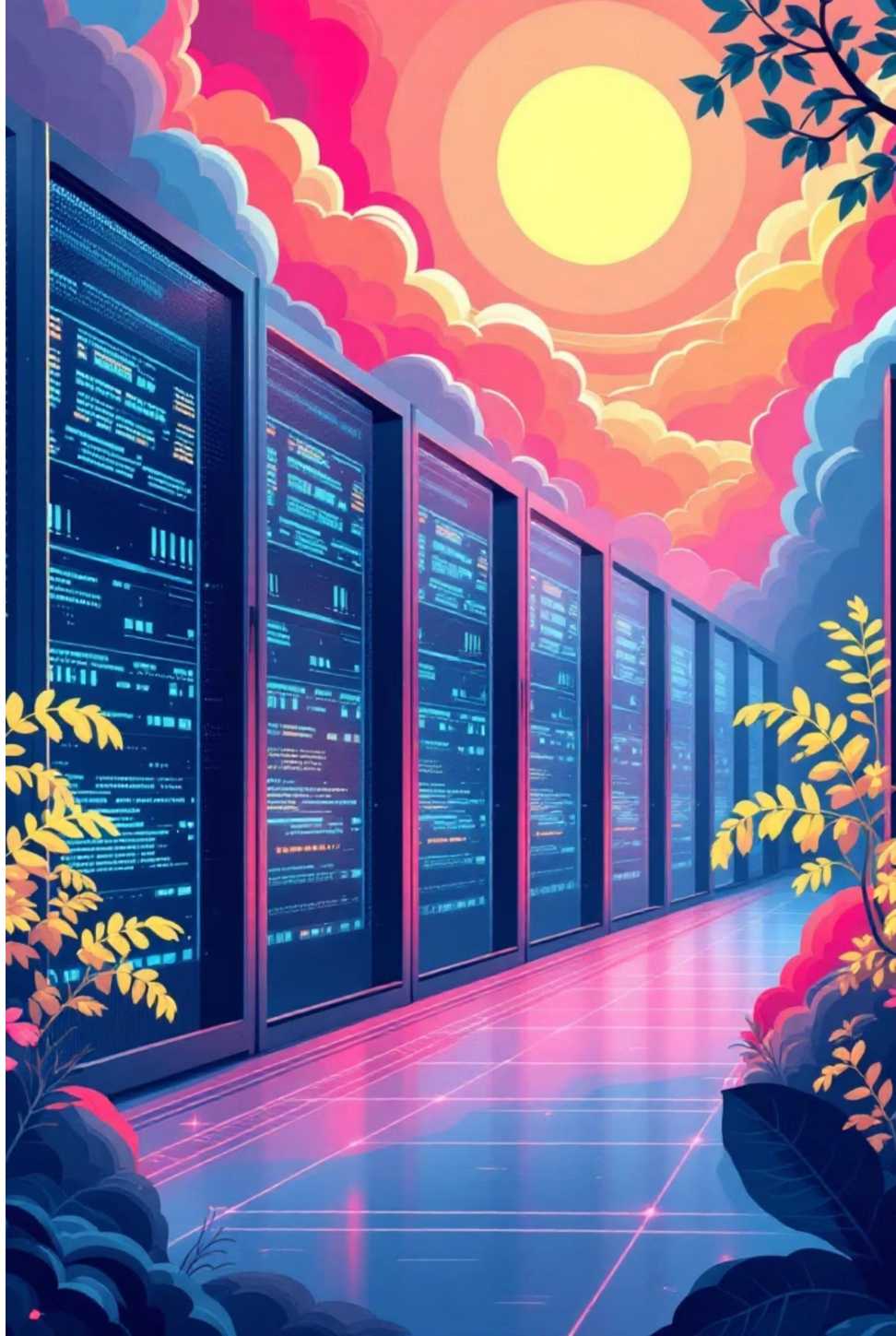
System Requirements

Functional System Requirements

- Handle user authentication securely
- Translate words using Prolog-based dictionary and Google API fallback
- Generate quizzes dynamically
- Store user progress and quiz scores
- Save audio recordings for pronunciation practice

Non-Functional System Requirements

- Respond quickly to user actions
- Support Sinhala and Tamil Unicode characters
- Handle multiple concurrent users
- Be scalable and deployable online
- Maintain data integrity and security





System Architecture: Layered Design

IruBasha employs a 4-Layer Client-Server Architecture, ensuring clear separation of responsibilities and adherence to industry-standard practices.

Presentation Layer

HTML, CSS, JavaScript for UI and user interactions.

Application Layer

Node.js with Express for API endpoints, authentication, and communication.

Logic & Intelligence Layer

SWI-Prolog for rule-based translations, vocabulary reasoning, and quiz generation.

Data Layer

SQLite for user accounts, quiz scores, word banks, and audio recordings.

Why Prolog for Logic & Intelligence?

Prolog, a logic programming language, is ideal for IruBasha's core functionalities, offering robust support for knowledge bases and deterministic rules.

Logic Programming

Efficiently handles complex linguistic rules.

Knowledge Bases

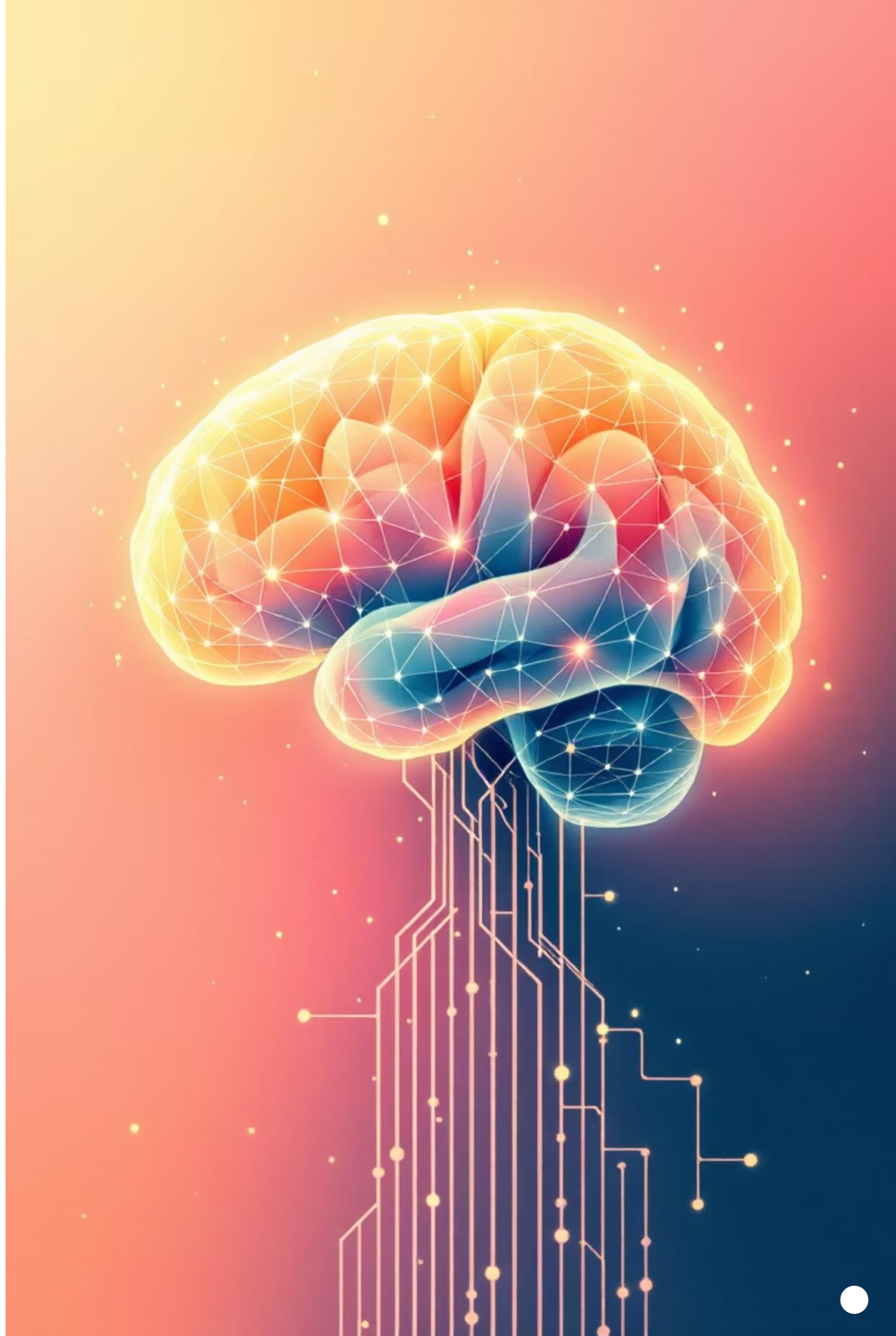
Perfect for managing and querying vocabulary data.

Deterministic Rules

Ensures consistent and accurate translations and quiz logic.

Educational Reasoning

Supports intelligent quiz generation and answer validation.



Live Demonstration Flow

The demo showcases IruBasha's error-free performance, correct AI integration, and user-friendly learning experience.

01

Log in / Registration

Secure access to the platform.

02

Translation

Word entry, Prolog/Google translation, text-to-speech.

03

Pronunciation

Record, save, and replay audio.

04

Quiz

Generate question, submit answer, view score and streak.

05

Word Bank

Save vocabulary, view mastery level.

06

Statistics

Progress overview, accuracy, and leaderboard.



Conclusion & Future Improvements

IruBasha successfully addresses early language learning challenges for Sri Lankan children with localized Sinhala–Tamil support, accurate Prolog-based translations, interactive pronunciation, and gamified quizzes.

Key Achievements

- Addressed early language learning challenges
- Provided localized Sinhala–Tamil support
- Implemented accurate Prolog-based translations
- Interactive pronunciation practice
- Gamified quizzes and progress tracking

Future Improvements

- Expand Prolog dictionary and sentence structures
- Add parent/teacher monitoring dashboard
- Introduce offline support using PWA
- Improve pronunciation feedback with speech analysis
- Deploy for real classroom usage in Sri Lankan schools